Computer Networks (EE-347)

Assignment 1: Client/Server Application using Sockets

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Server Code (server.py)

```
import socket
from datetime import datetime

HOST, PORT = "0.0.0.0", 5000

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
server.bind((HOST, PORT))
server.listen()
print(f"Server listening on port {PORT}")

count = 0
while True:
    conn, (ip, port) = server.accept()
    with conn:
        try:
            data = conn.recv(1024).decode()
            name, roll = data.split("|", 1)
            count += 1
            now = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
            reply = (f"Hello {name} ({roll})\n"
                     f"Your address: {ip}:{port}\n"
                     f"Request number: {count}\n"
                     f"Server date/time: {now}")
            conn.sendall(reply.encode())
            print(f"Request {count} from {ip}:{port} - {name} ({roll}) at {now}")
        except (ValueError, OSError) as e:
            print(f"Bad request from {ip}:{port}: {e}")
```

Client Code (client.py)

```
import socket
import sys

SERVER = sys.argv[1] if len(sys.argv) > 1 else "127.0.0.1"
PORT = 5000

name = input("Enter name: ")
roll = input("Enter roll number: ")

try:
    with socket.create_connection((SERVER, PORT), timeout=10) as s:
        s.sendall(f"{name}|{roll}".encode())
        reply = s.recv(1024).decode()
```

```
    print("\nServer says:\n" + reply)
except OSError as e:
    print(f"Could not reach server {SERVER}:{PORT}: {e}")
```

Helper (friend.py)

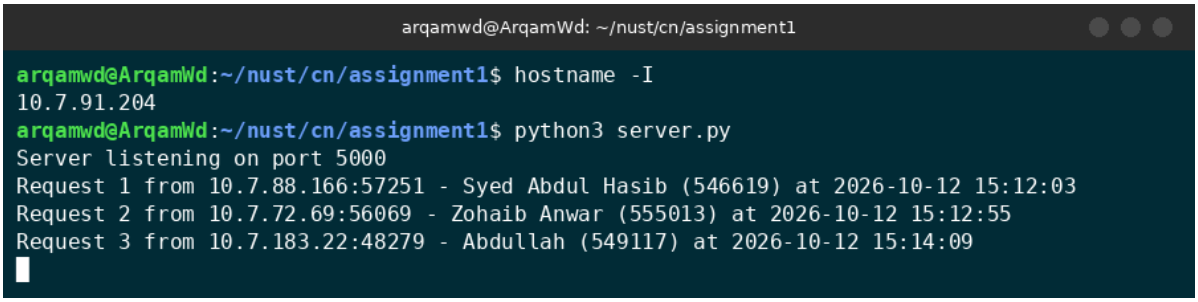
```
import runpy
import socket
import sys

ip, port = sys.argv.pop(1), int(sys.argv.pop(1))
_connect = socket.socket.connect

def connect(self, addr):
    self.bind((ip, port))
    return _connect(self, addr)

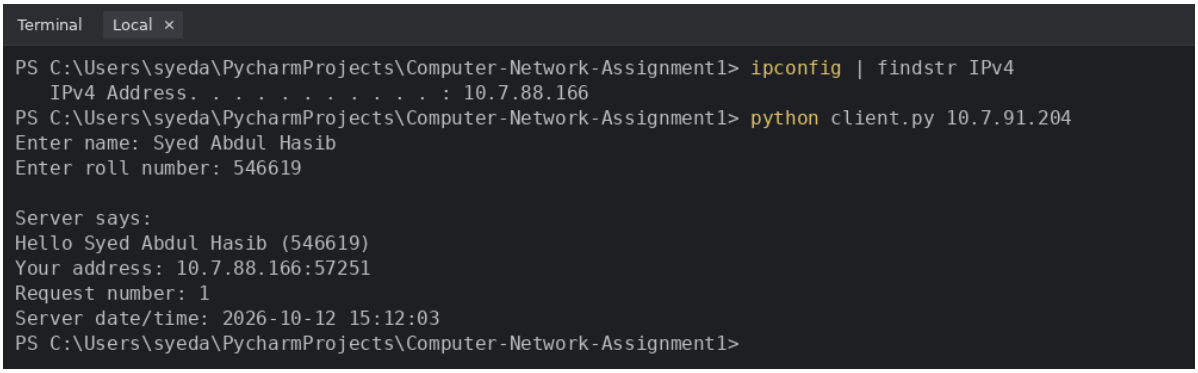
socket.socket.connect = connect
runpy.run_path("client.py", run_name="__main__")
```

Server Output

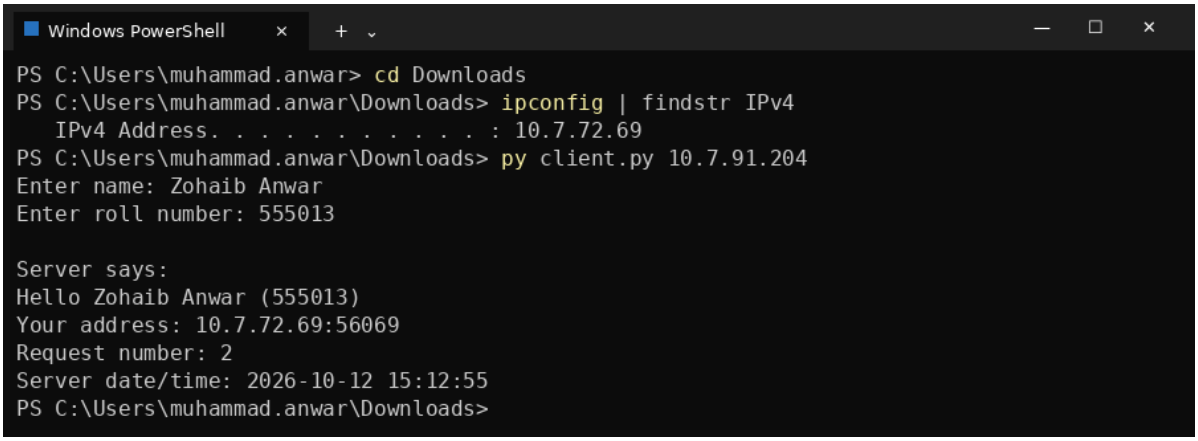


Server receiving 3 requests

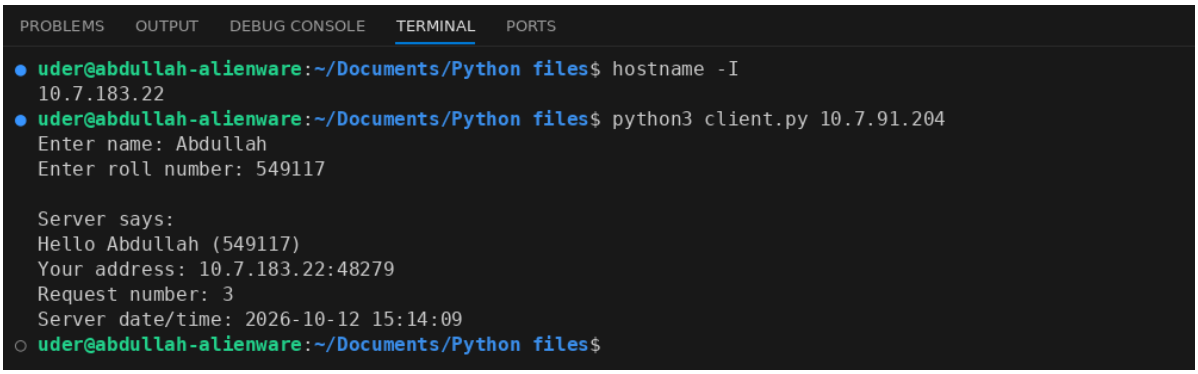
Client Outputs



Hasib (10.7.88.166)



Zohaib (10.7.72.69)



Abdullah (10.7.183.22)